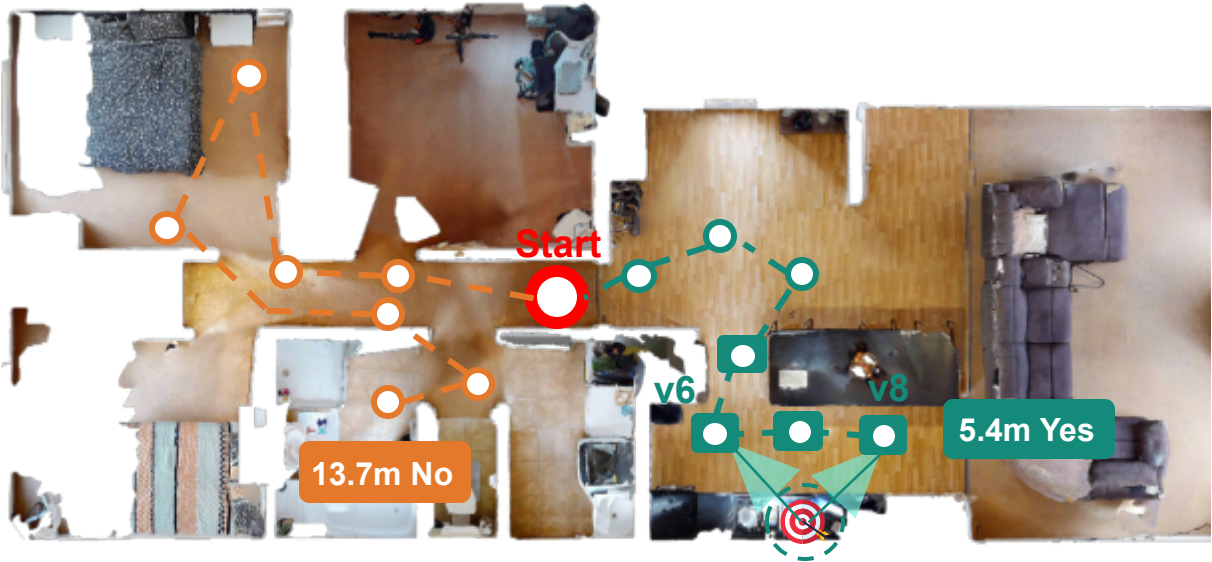


Query: “Is there a paper towel in the room?”

Same start · 8-action budget · ground truth: Yes



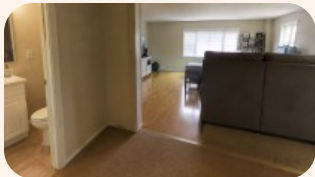
- Target
- Supporting view
- Category-room prior
- SafeVantage
- Estimate position

Category-room Prior

A room-prior exploration baseline

Start

initial set



score

0.00

Step 3

room prior - bedroom



score

0.00

Step 7

second sweep - bathroom

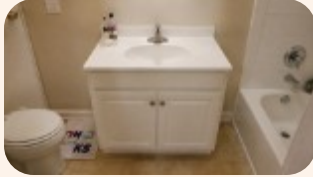


score

0.00

Step 8

budget exhausted



score

0.00

Evidence State:

support (“paper towel”) = 0

support set = $\emptyset$



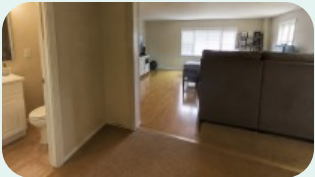
No

SafeVantage (Ours)

Vantage-aware memory drives selection

Start

initial set



score

0.00

Step 5

partial detection



score

0.31

Step 6

full detection



score

0.81

Step 8

corroborating view



score

0.95

exploration stage

corroboration stage

Memory:

support (“paper towel”) = 4

support set = {v5, v6, v7, v8}



Yes